

Relax! This test is not nearly as important as the start of Game of Thrones in nine days!

Each question is worth 2 points except where noted. Good luck!

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

Aquaporin	Hypertonic	Peristalsis
Amylase	Hypotonic	Potential
Bile	Iron bacteria	Sodium potassium pump
Chitin	Isotonic	Sphincter
Cellulose	Kinetic	Stomata
Enzymes	Methanogens	Sulfur bacteria
Flame bulb	Nephridiopore	Vitamins
Glycogen	Nephridostome	Waxy cuticle
Guard cells	Pancreas	
Hydrochannel	Pepsin	

1. **Aquaporin** Protein channel that increases water movement into cells.
2. **Amylase** The first digestive enzyme secreted by most animals during the digestive process.
3. **Kinetic** The energy of motion.
4. **Enzymes** Cofactors enhance the reaction ability of these macromolecules.
5. **Chitin** This structural molecule is found in cell walls and crab shells.
6. **Waxy Cuticle** Prevents water evaporation from most of the leaf surface.
7. **Hypertonic** The tonicity of the environment when Ψ of the environment is lower than Ψ in the organism.
8. **Nephridostome** Opening through which body fluids enter the metanephridia.
9. **ANY ANSWER** Found in the digestive tract of humans.

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

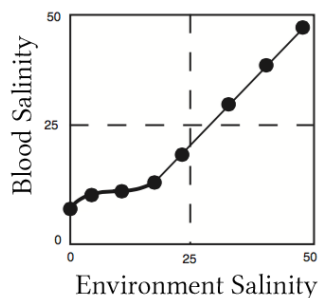
10. Upon activation by stomach acidity, the secretions of the gastric glands
- A. initiate the mechanical digestion of lipids in the stomach.
 - B. include nitrogenase and amylase.
 - C. delay digestion until the food arrives in the small intestine.
 - D. initiate the chemical digestion of lipids in the stomach.
 - E. initiate the digestion of protein in the stomach.**
11. Stomach cells are moderately well adapted to the acidity and digesting activities in the stomach by having
- A. secretions enter the stomach from the pancreas.
 - B. a high level of secretion of pepsin.
 - C. a thick, mucous secretion and active replacement of epithelial cells.**
 - D. a sufficient colony of *H. pylori* to breakdown the cellulose from vegetables.
 - E. a high level of bile.
12. Because the foods eaten by animals are often composed largely of macromolecules, this requires the animals to have mechanisms for _____ not found in other groups.
- A. elimination of nitrogenous waste.
 - B. reabsorption of solutes from the loop of Henle.
 - C. absorption of nutrients.
 - D. filtration in Bowman's capsule.
 - E. chemical digestion.**
13. Which of the following contains digestive enzymes?
- A. contractile vacuole.
 - B. food vacuole.
 - C. lysosome**
 - D. guard cells.
 - E. central vacuole.
14. Small mammals like mice use more energy per unit mass than large mammals mostly because most of the energy is used for
- A. the basal metabolic rate.
 - B. growing quickly.
 - C. activities like hunting.
 - D. thermoregulation.**
 - E. reproduction.
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15. Which of the following organs is incorrectly paired with its function?
- A. Mouth-starch digestion.
 - B. Small intestine-nutrient absorption.
 - C. Pancreas-enzyme production.
 - D. Stomach-protein digestion.
 - E. Large intestine-bile production.**
16. Which of the following statements is correct about diffusion in or out of a cell?
- A. It is an active process in which molecules move from a region of lower concentration to one of higher concentration.
 - B. It requires channel proteins in the cell membrane.
 - C. It is a passive process in which molecules move from a region of higher concentration to a region of lower concentration.**
 - D. It requires aquaporins in the cell wall.
 - E. It requires an expenditure of energy by the cell.
17. Ingested dietary substances must cross cell membranes to be used by the body, a process known as
- A. ingestion.
 - B. hydrolysis.
 - C. absorption.**
 - D. elimination.
 - E. digestion.
18. When the digestion and absorption of carbohydrates results in more energy-rich molecules than are immediately required by an animal, the excess is
- A. stored as glycogen or fat.**
 - B. eliminated in the feces.
 - C. stored as starch.
 - D. stored as cellulose or chitin.
 - E. stored in the crop.
19. Stomata will open when
- A. the guard cells have a lower solute concentration than outside the cell.
 - B. the water potential inside the guard cell is higher than outside the cell.
 - C. the solute potential inside the guard cell is positive.
 - D. the pressure potential inside the guard cell is negative.
 - E. the guard cells have a higher solute concentration than outside the cell.**
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20. A marine iguana is basking on a sunny rock after feeding in the cold Pacific Ocean water. It's warming its body through
- A. conduction and convection
 - B. convection and radiation.
 - C. conduction and evaporation.
 - D. convection and evaporation.
 - E. conduction and radiation.**
21. Organisms that obtain energy by oxidizing iron compounds and obtain carbon from CO_2 are called
- A. chemoheterotrophs.
 - B. chemoautotrophs.**
 - C. photochemotrophs.
 - D. photoautotrophs.
 - E. photoheterotrophs.
 - F. autoheterotrophs.
22. The advantage of excreting nitrogenous wastes as urea rather than as ammonia is that
- A. urea can be exchanged for Na^+ .
 - B. urea does not affect the osmoregulation.
 - C. urea requires more water for excretion than ammonia.
 - D. less nitrogen is removed from the body.
 - E. urea is less toxic than ammonia.**
23. Which of the following is an accurate statement about thermoregulation?
- A. Endothermy has a lower energy cost than ectothermy.
 - B. Endotherms maintain a constant body temperature and ectotherms do not.
 - C. Endotherms are regulators and ectotherms are conformers.
 - D. Endotherms and ectotherms differ in their primary source of heat for thermoregulation.**
 - E. Endotherms are warm-blooded and ectotherms are cold-blooded.
24. The osmoregulatory/excretory system of a freshwater flatworm is based on the operation of
- A. protonephridia.**
 - B. metanephridia.
 - C. nephrons.
 - D. ananephredia.
 - E. Malpighian tubules.
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25. Pepsin is a digestive enzyme that
 - A. present in bile to digest lipids in the small intestine.
 - B. digests polysaccharides to monosaccharides in the small intestine.
 - C. digests nucleic acids in the stomach.
 - D. begins the digestion of proteins in the stomach.**
 - E. secreted by the pancreas to digest proteins in the duodenum.
 26. A fungus is a type of
 - A. autoheterotroph
 - B. chemoautotroph
 - C. chemoheterotroph**
 - D. photoheterotroph
 - E. photoautotroph
 27. Which of the following types of molecules are the major structural components of the cell membrane?
 - A. phospholipids and proteins.**
 - B. proteins and collagen.
 - C. proteins and cellulose.
 - D. nucleic acids and proteins.
 - E. phospholipids and cellulose.
 28. After ingestion by humans, the first type of macromolecules to be chemically digested by enzymes in the mouth is
 - A. carbohydrates.**
 - B. proteins.
 - C. nucleic acids.
 - D. minerals.
 - E. cholesterol and other lipids.
 29. The transfer of fluid from the glomerulus to Bowman's capsule
 - A. is very selective as to which protein-sized molecules are transferred.
 - B. usually includes the transfer of red blood cells into Bowman's capsule.
 - C. results from active transport.
 - D. transfers large molecules as easily as small ones.
 - E. is mainly a consequence of blood pressure in the capillaries of the glomerulus.**
 30. A species of rabbit that follows Bergmann's rule
 - A. would have larger ears in Maine and smaller ears in Georgia.
 - B. would be whiter in winter and browner in summer.
 - C. would be larger in Canada than in Texas.**
 - D. would be smaller in Minnesota than in Florida.
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31. Cell walls are found in at least some representatives of
- A. bacteria
 - B. plants
 - C. all of the above.**
 - D. algae
 - E. fungi
 - F. protists
32. Which of the following membrane activities requires energy from ATP?
- A. movement of glucose molecules into a bacterial cell from a solution containing a higher concentration of glucose than inside the cell.
 - B. movement of water into a cell.
 - C. diffusion of chloride ions across the membrane through a chloride channel.
 - D. movement of Na^+ ions from a lower concentration in a mammalian cell to a higher concentration in the extracellular fluid.**
 - E. movement of water out of a paramecium.
33. Birds secrete uric acid as their nitrogenous waste because uric acid
- A. is metabolically less expensive to synthesize than other excretory products.
 - B. excretion allows birds to live in desert environments.
 - C. is less toxic than urea.**
 - D. is readily soluble in water.
 - E. requires little water for nitrogenous waste disposal, thus reducing body mass.**
34. Which of the following is *not* a major activity of the stomach?
- A. Mechanical digestion.
 - B. Mucus secretion.
 - C. Nutrient absorption.**
 - D. HCl secretion.
 - E. Enzyme secretion.
35. The following figure shows the blood salinity level of a marine worm compared to the salinity of the surrounding environment. Based on this figure this organism is most likely an

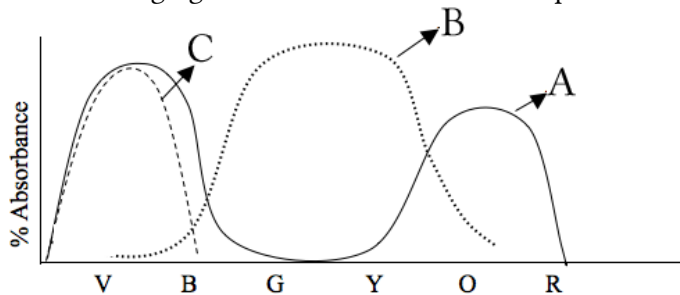


Osmoconformer

Circle True or False. Read each statement completely before answering.

36. True **FALSE** The sources of metabolic nitrogen in the diet are proteins and carbohydrates.
37. True **FALSE** Micronutrients are smaller molecules than macronutrients.
38. **TRUE** False The chemical energy stored in carbohydrates is a form of potential energy.
39. True **FALSE** A euryhaline species cannot survive in a hypotonic environment.
40. **TRUE** False A flaccid plant cell is most likely to occur in an isotonic environment.
41. True **FALSE** Cation exchange occurs when plants release CO_2 into the soil to increase the acidity.
42. True **FALSE** Of the six organismal groups we are comparing, only the bacteria and the protists are not monophyletic.
43. **TRUE** False All organisms are considered open systems.
44. **TRUE** False The primary function of the large intestine is the absorption of water.
45. True **FALSE** The central vacuole is used by protists to eliminate excess water from the cell.

Use the following figure to answer the next three questions.



46. Line A represents the absorption spectrum of which group of photopigments? **chlorophyll**
47. Line B represents the absorption spectrum of which group of photopigments? **phycobilins**
48. Line C represents the absorption spectrum of which group of photopigments? **carotenoids**
49. (4 points) Write a *balanced* chemical equation for photosynthesis. $6 \text{CO}_2 + 6 \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{O}_2 + 6 \text{H}_2\text{O}$
Subtract 1 pt if one prefix number is missing or wrong (e.g., 4 H_2O , 2 pts if more than 1 is wrong.
Subtract 1 pt if one chemical formula is wrong (e.g., $\text{C}_{12}\text{H}_6\text{O}_6$), 2 pts if one missing more more than 1 wrong.
Ask me if uncertain.